



Folkhälsomyndigheten

Dealing with the health effects of heat waves

Guidance for action plans



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About the publication

The Public Health Agency works preventively around the negative impact of heat waves on health.

This guidance can provide support and recommendations to municipalities, regions and private actors in their work to develop action plans to strengthen the ability to manage negative health effects.

The action plans can contribute to reducing the population's and especially the vulnerable groups' risk of dying or falling ill due to heat waves.

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Content

Summary.....	9
Heat waves in Sweden	11
Health effects of heat	11
Risk groups.....	12
Preparedness for heat waves.....	12
Warning system for high temperatures in Sweden	13
Developing an action plan.....	15
Find the appropriate level of ambition - and a basis for systematic improvement.....	15
Prepare – plan the communication, train and practice	15
Manage – gain time with early warning and alarm chain	22
Reset	24
Learning - important to document and evaluate	25
Prevent – on measures and responsibilities to reduce people's exposure to heat.....	27
Knowledge support to reduce health-damaging heat	27
Method for mapping risk areas in existing buildings	27
Preventive work in spatial planning	28
Liability of the property owner	28
Requirements in procurement and contracts with private contractors.....	28
Supervision and aid processing	29
References.....	30
Appendix 1. Example of a news release or press release	32
Appendix 2. Examples of instructions for various functions in the alarm chain	33

Summary

A widespread heat wave can have serious consequences for society and not least for the population's health. Therefore, the World Health Organization (WHO) recommends that countries in Europe develop and implement plans to prevent and manage heat-related health consequences.

This guidance will provide support and recommendations for organizations' work in developing action plans to prevent and manage negative health effects of heat waves. The guidance is written for municipalities, regions and private actors who are responsible for care and care and provides information to the groups that are vulnerable to the negative health effects of heat waves.

The negative effects of heat on health cover a broad spectrum from relatively mild symptoms such as dehydration and reduced general condition, to more serious symptoms such as heat stroke and heart attack. Being exposed to heat entails different risks for different individuals. **Risk groups are above all the elderly, chronically ill, people with disabilities, young children, pregnant women and people taking certain medications.**

Action plans are a way to ensure that organizations work systematically to manage health risks from heat waves. An action plan for heat waves should contain measures that strengthen the ability to reduce negative health effects and that strengthen the population's resilience, as well as have documented routines for prevention, preparation, handling, restoration and learning.

The guidance provides examples of how to work to develop action plans as well as important aspects to consider in the development. An alarm chain can be set up in order to reach the care operations as quickly as possible and without delay pass on the heat warning as well as advice adapted to the target group on how to avoid health effects.

As support for the work with action plans, the Public Health Agency has produced information material that contains advice adapted to the target group and that can be used by municipalities, regions and private actors in their work to reduce the health risks linked to heat waves. The material can be downloaded from www.folkhalsomyndigheten.se/varmebolja

The earlier in the causal chain that the negative health effects of heat waves can be dealt with, the greater the potential for health gains and the less the need for preparedness measures. In the chapter Prevention - on measures and responsibilities to reduce people's exposure to heat, there are examples of preventive measures that can be implemented around heat.



Heat waves in Sweden

Global warming has caused the climate to change and heat waves can no longer be considered something that rarely occurs. In Sweden, such temperatures, which currently occur every twenty years on average, may occur every three to five years by the end of this century (1).

A widespread heat wave can have serious consequences for society and not least for the population's health. Therefore, the World Health Organization (WHO) recommends that countries in Europe develop and implement plans to prevent and manage heat-related health consequences (2). It is important that decisions on measures are taken in advance and that these are followed in the event of a heat wave. These also need to be communicated to relevant actors (3). The need for coordinated recommendations has been highlighted as urgent by several actors in Sweden (4-6). The authority for social security and preparedness (MSB) has previously worked on producing information sheets in various sectors for municipalities (7).

Sweden has a relatively cold climate, so heat is something positive for the vast majority of people. Therefore, people like to go out and sun exposure, also in healthcare. This can lead to trivializing or ignoring the problem, especially if there is a lack of knowledge that the risks of heat can be complex based on individual health conditions and medication, and that symptoms of heat stress can be diffuse and insidious.

This guidance should provide support and recommendations for the work of various actors in developing action plans to prevent and manage negative health effects of heat waves. The guidance is written for municipalities, regions and private actors who provide health care to the population, and especially the risk groups that are sensitive to the negative health effects of heat waves.

Compilation of knowledge on the health effects of heat waves

The basis for this guidance is, among other things, a compilation of knowledge about the health effects of heat waves (8). The summary describes the state of research regarding risk groups and the health effects of heat waves as well as preventive management in society.



Health effects of heat

The large number of deaths that occurred during the heat wave in Europe in 2003 showed the consequences for public health. Up to 70,000 more deaths than normal are estimated to have occurred and mortality increased at all ages (5). In the summer of 2018, Sweden was hit by a long-lasting heat wave with record high temperatures across the country. The Public Health Authority assessed the excess mortality at approximately 700 cases for the entire summer.

During heat, the temperature in the body is regulated partly by increasing blood circulation in the superficial blood vessels and dilating the vessels, partly by increasing sweating. This regulation in turn leads to fluid and salt losses and a greater strain on the heart (8).

The negative health effects of heat usually occur soon after high temperatures are noted, often already on the same day or after one to two days. The negative effects of heat on health cover a broad spectrum from relatively mild symptoms such as dehydration and reduced general condition, to more serious symptoms such as heat stroke and heart attack (8).

Risk groups

Being exposed to heat entails different risks for different individuals (9). This applies above all (8, 10):

- **Elderly.** Older people have poorer temperature regulation than younger people. They may have a reduced ability to feel thirst, and more often have chronic diseases. Some medications that are common among the elderly can cause serious side effects at high temperatures. Heat-related deaths in the elderly are usually due to a worsening of an existing underlying disease or circulatory disorders. Living alone, being bedridden or not being able to get out on your own is also associated with increased mortality during a heat wave.
- **Chronically ill.** Common chronic diseases such as cardiovascular disease, asthma and COPD is greatly aggravated by heat. Kidney disease, dementia, diabetes and mental illness increase the risk of health problems as a result of heat waves.
- **People with disabilities.** In the case of a mental disability, it can be more difficult to perceive risks and interpret the body's warning signals. Even in the case of physical impairment, the body's signaling system can be out of order. People with physical disabilities may need practical help.
- **Small children and pregnant women** can be particularly sensitive when it gets hot. Small children have usually not fully developed their ability to sweat. Very young children are usually unable to move themselves away from the heat and are therefore dependent on the help of adults. Pregnant women have a certain risk of giving birth prematurely.
- **People taking certain medications.** Some medications affect the body's adaptation of circulation and fluid balance to high temperatures. This applies, among other things, to diuretics or antidepressants and neuroleptics.

Preparedness for heat waves

Preparedness work can be divided into different phases in order to achieve systematic work (11). In this guidance, the phases are named: prevent, prepare, manage, restore and learn, see figure 1 (12).

Figure 1. Different phases in the preparedness work.



In the preventive phase, measures are implemented early in the causal chain to reduce the risk of various contingency situations arising. Acute events are counteracted and prevented already at the source. In the chapter Prevention - on measures and responsibilities to reduce people's exposure to heat, there are examples of preventive measures that can be implemented around heat and thus reduce the risk of negative health effects occurring.

Action plans for heat waves are developed primarily in the preparatory phase of preparedness work. The action plans aim to ensure that the negative health effects that can be expected to occur during heat waves are handled in an efficient and systematic way. The preparatory phase is mainly about trying to prepare routines, organization and instructions in advance in order to thereby strengthen the ability to handle emerging situations effectively.

In the handling and restoration phases, you act based on the routines and the organization that have been worked out. Then it is a matter of handling the heat wave to the best of our ability and in accordance with the principles that prevail for the organisation's and society's preparedness work. When the acute handling phase has ended, action is taken to restore the business again.

A learning phase provides a fundamental value for all preparedness work. Through follow-up of events and an evaluation of measures and action plan, there is an opportunity to further develop and adapt preparedness.

Warning system for high temperatures in Sweden

Since 2014, SMHI has been warning of high temperatures. SMHI sends out:

- Notification of high temperatures when the highest temperature of the day is expected to be between 26 and 30 degrees Celsius for three consecutive days.
- Yellow warning when the daily maximum temperature is expected to reach 30 degrees Celsius or more for three to four consecutive days.
- Orange warning when the highest temperature of the day is expected to reach 30 degrees Celsius for five consecutive days and/or when the highest temperature of the day is expected to reach 33 degrees Celsius for at least three consecutive days.

SMHI spreads messages and warnings via smhi.se, the [SMHI Väder](#) app and Sweden's Radio P1 and P4. The same information is also available via krisinformation.se. [More information about the warning system](#) is available on SMHI's website.



To develop an action plan

Action plans for heat waves are a way to ensure that organizations work systematically to manage health risks from heat waves. An action plan for heat waves should contain measures that strengthen the ability to reduce negative health effects.

The measures that society has prepared and that are taken in case of heat warnings determine the extent to which the heat wave affects public health. The measures can differ depending on whether they are aimed at the societal level or at the individual level. The measures can also be designed differently depending on which conditions apply within an organization and which target groups are prioritized (13).

Find the appropriate level of ambition – and a basis for systematic improvement

Designing a preparedness for heat waves should be done with a level of ambition that is appropriate based on the organization's conditions and the resources that are available.

An action plan for heat waves can form part of a more ambitious preparedness plan that considers and prepares the entire organization for various emergency situations.

Action plans can also be designed in a simpler form and then cover parts of the operations in a municipality, a region or with a private actor. This can, for example, consist of the municipality's health and care administration having developed advice that is conveyed to staff and risk groups in the event of a heat wave.

Regardless of delineation and level of ambition, the action plan should have documented routines for prevention, preparation, management, recovery and learning.

Prepare – plan the communication, train and practice

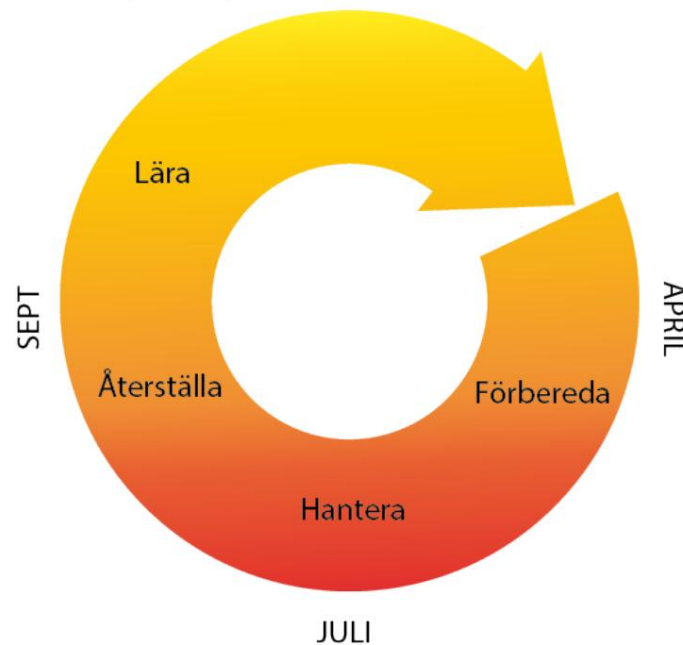


In many cases, there is neither mandate, time nor, for other reasons, the opportunity to make preparatory decisions during the current heat wave. The work on preparedness for dealing with a heat wave therefore needs to start well in advance of the summer.

Information and training regarding action plans for heat waves as well as decided measures should go out widely in the business in time, not least to summer staff, but also to, for example, communicators and IT managers who can then prepare and test documents and routines for their respective efforts. In the preparatory phase, decisions can be made within the regular chain of responsibility.

Work on preparedness for heat waves extends throughout the year. Figure 2 describes important phases in developing action plans and working strategically with heat waves, as well as when during the year the work in the various phases has its center of gravity.

Figure 2. Annual cycle of important phases in the work of developing action plans for handling in the event of a heat wave. The preventive phase is worked on throughout the year, and long-term.



Plan the communication and prepare communication support

Many are affected in the work of developing action plans for heat waves, and cooperation is required to find effective communication channels for the decided measures.

Communication functions and communication managers within municipalities, regions and private actors should be involved early in the work because they play an important role in the implementation and in the design of various communication supports. It is important to consider the communication challenge that arises in heat wave preparedness. It can be difficult to get across with messages about negative health effects. Heat is often seen as something that is essentially positive and this, in combination with a lack of knowledge about health effects, can mean that target groups may trivialize or not take on board the problem.

EXAMPLE OF COMMUNICATOR SUPPORT

- Identify possible information channels based on target groups.
- Prepare templates for news releases and press releases, see examples in Appendix 1.
- Prepare function emails in the preparatory phase to facilitate the spread of the heat warning.
- Create a routine for administrators who serve during the summer.
- Identify technical communication support, for example rolling banner in the patient record systems, different types of crisis apps and SMHI's weather app.

Identify and define target groups

Messages about the negative health effects of heat waves need to be formulated in a way that clearly reaches the target groups. Adapting the communication to the target group can improve the possibilities of achieving an effective alarm chain and good results with the decided measures in the action plan. It is therefore important to clearly identify and define your target groups for the action plan.

Target groups in the alarm chain

- Managers with responsibility for crisis preparedness and health and care
- Security and preparedness coordinators as well as officials in health and social care
- Officer on standby (TiB)
- Local and regional cooperation groups within preparedness •
Operations managers
- Medically responsible nurse (MAS)
- Legitimate healthcare staff
- Communications manager

It is important that the target groups that are to act in the alarm chain have the same situational picture and know their function in case of heat warnings. Knowledge and routines need to be developed and incorporated and repeated in advance, so that decided measures can then be put into practice in the event of an alarm or notification of a heat wave.

Target group: healthcare professionals

- Doctors
- Nurses
- Nurses
- Care assistants
- MVC nurses
- BVC nurses
- Responsible for food and purchases

The healthcare staff's needs linked to heat waves mainly consist of receiving information that heat warnings are in place and that special routines have been activated, what they should be aware of regarding the health of risk groups and what preventive advice they can give to risk groups. Therefore, it is important that the action plans are filled with measures also in the form of concrete advice and information material that can be communicated to the risk groups in advance and during a heat wave.

The operations managers' function and participation in the alarm chain is of crucial importance for healthcare personnel to be able to act according to the decided measures. Operations managers are important intermediaries in the alarm chain. It is important that operational managers and also nursing staff are included in exercises carried out in the alarm chain.

Having access to the right food and drink is important during a heat wave. Those responsible for food handling and purchasing therefore need to be notified early that a heat wave is on its way and that they have the knowledge and preparedness to make well-thought-out purchases and adaptations. This may involve changing the menu to more liquid-rich food and vegetables or adapting routines for storage and cooling. The care staff may also need access to fruit, ice, cold drinks and ice cream that can be served between meals.

Target group: risk groups who have contact with healthcare

The healthcare staff can play an important role in the health development of the individuals who are in regular contact with healthcare. Information and advice about heat waves can therefore be advantageously conveyed in a coordinated and efficient manner via healthcare personnel.

Target group: risk groups who do not have contact with healthcare

A relatively large proportion of the individuals in the risk groups are outside the care chain and are not in contact with care, especially in the risk group "elderly". For example, in the municipality of Uppsala in 2014, 87% of people older than 65 did not have a place in accommodation or home care. For people over 80, the figure was 65% and for people over 90, 35% had no regular contact with healthcare (14).

It is a communicative challenge to reach people who belong to the risk groups but are not under care or have regular contact with care. The communication needs to be designed so that it reaches people who do not have knowledge of the risks that heat can entail and for other reasons may not perceive themselves as belonging to a risk group.

To reach this group, special measures and information channels should be identified.

Inform and educate

Education and information play a big role in how well the alarm chain will work in the event of a heat wave and how information about advice reaches the target groups. It is therefore important in the preparatory phase to inform and educate within the organization about health effects, the decided action plan and measures to deal with the situation.

The implementation can take place through, for example, business planning, workplace meetings, staff days, training opportunities or exercises. Websites and tools and forums for internal communication are also important channels for this. Information via websites, media, pharmacies, 1177, interest organizations and relatives can play an important role for risk groups, and especially people without healthcare contact.

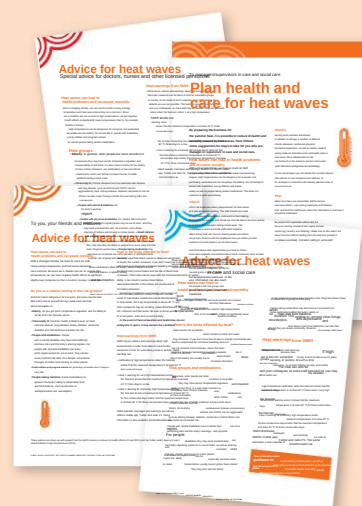
Information material tailored to the target group is available as support

As support in the work, the Public Health Agency has produced information material that can be used by municipalities, regions and private actors in their work to reduce the health risks linked to heat waves. The material can be downloaded from www.folkhalsomyndigheten.se/varmebolja

The advice given by various actors to minimize morbidity and mortality at high temperatures and heat waves is consistent. The information material is designed with the support of the WHO's advice to reduce the health effects of heat and the knowledge compilation Health effects of heat waves as well as text material from Klimatsamverkan Skåne's preparedness plan for heat waves (6, 8, 15).

Information material

Advice on what to think about and how to act is available to • managers and supervisors for special accommodation, home health care and in primary care • doctors, nurses and other authorized persons personal • care and social care staff in home care, home health care, special housing and primary care • you, your friends and relatives.



Two short brochures with information about risks and practical advice during heat waves have been produced.

Dealing with heat waves - information about risks and practical advice for staff in care and social care

When it gets too hot - advice for you, your friends and relatives during a heat wave.

Animated films



Advice during heat waves - for you, your friends and relatives



Advice in case of heat waves - for health and care workers

Web education



An online course has been developed which is aimed at healthcare professionals *Managing heat waves in healthcare – test what you can and learn more!*

Image bank



An image bank of images that can be used to illustrate different advice during heat waves

Practice – to test routines and maintain knowledge

A fundamental reason for practicing is to give different target groups the opportunity to meet and work together across administrative boundaries. These established contacts can be of great benefit when something happens. It is also an advantage that roles and responsibilities are made clear through exercises (16).

Exercises can develop different types of abilities in the organization and the alarm chain, for example the ability to cooperate or the ability to create a common situational picture. Exercises can also, in a different way than self-assessments, test an organization or decided action plans. Furthermore, exercises can support the development of new abilities and improvements in routines, action plans and measures (16).

Heat waves are something that returns with a certain regularity and most people can relate to some heat wave and what it was like to work during such a period. Precisely because we know that many people will be affected by heat waves, it is important to practice in advance for such an event scenario. An event scenario for a heat wave has been developed by the county administrative boards and by MSB (17,18).

Inspection of premises and equipment

Important preparations are also to carry out an inspection of the premises in good time and to acquire equipment that facilitates and speeds up the possibility of reducing the health effects of a heat wave.

Review whether the care staff can have access to alternative premises, such as basement spaces or cool premises and patios where the care recipients can move. Also check whether windows that are exposed to direct sunlight have functioning sun protection, for example awnings and blinds, and whether the healthcare staff can access equipment such as indoor thermometers, portable air conditioning and ice machines.

Be judicious about the use of fans, as they can have a drying effect. It is still not clear under which conditions fans do more good than harm (8).

Manage - gain time with early warning and alarm chain



Heat waves can lead to health problems and increased mortality already during the first day of the heat wave. An alarm chain can be set up to pass on a heat warning from SMHI as quickly as possible and without delay, as well as advice adapted to the target group on how to avoid health effects.

Design of alarm chain

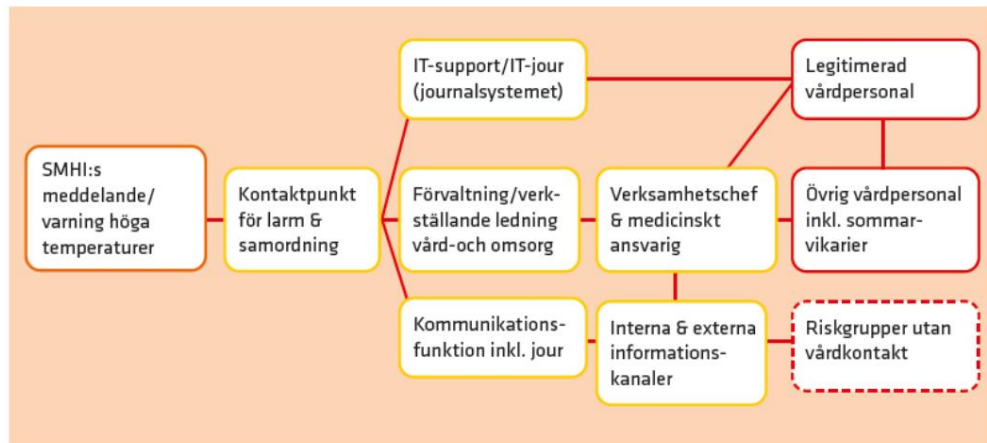
In the design of an alarm chain, there are many aspects that need to be taken into account, and therefore the alarm chain is advantageously drawn up early in the preparatory phase, in order to be quickly activated in the handling phase.

Aspects that need to be taken into account range from roles and responsibilities in the business to what obstacles and opportunities exist in the technical communication systems. The organization's internal management needs to be prepared, but also cooperation with other actors in society.

Important things to consider when designing the alarm chain:

1. In the handling phase, it is healthcare personnel and risk groups who primarily need be reached. The alarm chain should not be delayed by SMHI's message or warning about high temperatures entering too high in the organization. The more links in the alarm chain, the more vulnerable it is to delays.
2. It is good to secure with several alternative branches in the alarm chain, where heat warnings are conveyed partly through ordinary information channels and partly via faster information channels directly to healthcare personnel. This can be done via journal systems, apps, intranets and external webs, see figure 3 (alarm chain).
3. An alarm chain for handling heat waves needs to take into account holiday periods during which the regular chain of responsibility is often broken and the regular care staff are not on site.
4. It is important to consider the transition from office hours to evening and weekend hours. On-call functions are important parts of the alarm chain and the transition itself can be particularly vulnerable and therefore requires clear procedures.
5. Verbal direct contact is often most effective, but should preferably be combined with written contact information.

Figure 3. Schematic overview of possible functions when designing an alarm chain, and important recipients of SMHI's message or warning about high temperatures.



Design instructions that carry four important routines

By designing clear instructions for each function in the alarm chain, several important steps can be carried out and passed on at different levels in the organization. Examples of the design of instructions can be found in Appendix 2.

The instructions can let the alarm chain convey four important steps, see figure 4.

Figure 4. Four important moments that the alarm chain can convey.

Convey heat warning



The organization should designate a contact point for alarms and coordination internally within the organization, but also for cooperation with other actors. It can be an officer on standby (TiB), officer on duty, orientation and coordination contact (Sambruk) or equivalent.

Remember to design the instructions so that the alarm chain updates the heat warning if the level of the message or warning changes. The warning level may increase or decrease depending on the weather forecast and other factors.

Provide advice and information materials

Distribute target group-adapted instructions, advice and information material to the various functions in the alarm chain.

Disseminating advice and information material is best done in the preparatory phase, but should also be done in the event of a heat warning.

Feedback

Create a common situation picture by exchanging information with other functions in the alarm chain. It may be about feedback on what consequences the heat has had, what needs for help have arisen, what coordination is lacking, what resource availability is possible. This contributes to the fact that various functions in the business are involved and can contribute to the situation. It also enables an assessment of the consequences of the event in the business in the short and long term and can therefore contribute to insights and improvement in the face of future heat waves.

Document

Document contacts and activities. Here is an example documentation template:

Datum & tid	Rapportör	Händelse/aktivitet	Åtgärd/beslut	Status	Minnesanteckning

The organization should have procedures for documenting considerations and decisions before, during and after an event. The documentation is important to be able to follow up the work and it lays an important foundation for new knowledge, conclusions and learning, see the chapter Learning – important to document and learn.

Restore

FÖREBYGGA	FÖRBEREDA	HANTERA	ÅTERSTÄLLA	LÄRA
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The restorative phase is limited in the work with action plans for heat waves and mainly includes feedback in the alarm chain that the heat warning no longer applies.

However, it may be important to call for continued attention even after a heat wave because the health effects may persist or occur at lower temperatures than the threshold values where warnings are issued (8).

In addition, it is important to check that medicines have been stored according to the instructions on the packaging or in the package insert.

Learning - important to document and evaluate



Routines are needed to follow up and evaluate action plans. It is also important to make use of experiences from incidents and exercises to develop and refine the alarm chain, division of responsibilities, instructions and information materials.

Follow-ups should always be done in connection with the actual handling of a heat wave, but also at pre-defined times to systematically integrate new knowledge as well as changed organizational and technical conditions.

Examples of data that can be used for follow-ups are interviews, questionnaires and round table discussions with relevant target groups, summaries of experiences from exercises and logbook notes.



Prevent – about measures and responsibilities to reduce people's exposure to heat



In addition to knowledge-raising measures about health effects and management in health care, preventive measures are needed that reduce people's exposure to heat. The earlier in the causal chain that health risks from heat waves are addressed, the greater the chances of avoiding negative health effects and the less the need for handling preparedness measures.

A heat wave is, according to the Swedish Agency for Community Protection and Preparedness, a natural disaster and extraordinary event (19). Heat waves should therefore be included in the risk and vulnerability analyzes and management plans that according to law (LEH 2006:544) must be carried out in municipalities and regions.

Knowledge support to reduce health-damaging heat

The population may be more exposed to heat in urban areas than in sparsely built-up areas. This particularly applies to densely built-up areas with a high proportion of hard surfaces and little vegetation. In such areas, both air temperature and radiation temperature can become high, and there is a higher risk of so-called heat islands forming. The municipality has an important role in terms of increased knowledge of the problem, adapting street and park environments, and in its community building function steering towards climate adaptation, even in existing buildings.

The Public Health Agency has knowledge support for municipalities and property owners in their preventive action work. Knowledge support Heat and people in the built environment (20) compiles health risks and society's vulnerabilities to heat waves, factors in the built environment that can cause high temperatures, as well as physical measures that can be taken both outdoors and indoors to reduce the risk of heat that is harmful to health in existing buildings.

Method for mapping risk areas in existing buildings

The Norwegian Public Health Authority has developed a GIS method for mapping buildings that are at risk of developing high temperatures (21). The method is based on the geographic data on land cover that are readily available to municipalities and county administrative boards.

The method is described step by step and culminates in a map image that illustrates at a glance which areas in the urban area are at greater risk of developing health-damaging heat, and where measures need to be implemented first.

In healthcare and social care, the mapping of risk areas can be supplemented with the identification of where the risk groups live, and then become useful in the planning of care interventions, for example prioritization and visit frequency for home care and home care during a heat wave.

Preventive work in physical planning

Buildings and infrastructure built today will be around for a long time. It is therefore important to take future heat waves into account as much as possible when planning and building.

The Norwegian Public Health Agency has knowledge support that highlights the importance of planning and building health-promoting living environments (22). The knowledge support The importance of our living environment for good and equal health addresses, among other things, how climate change amplifies existing health problems, including heat waves, as well as the importance of greenery in the built environment (23).

In the PBL knowledge bank, the Housing Authority has guidance regarding, for example, heat waves, green plans and climate adaptation.

Liability of the property owner

Private property owners and public property management need to adapt buildings and outdoor environments to today's and tomorrow's heat waves.

According to law, both the property owner and the person who runs a business have a responsibility to ensure that the property is maintained and taken care of so that the residents do not suffer or become ill from defects in the building. Informing and conducting dialogue with the users within a property portfolio can contribute to more efficient measures and that buildings are used in the right way when the heat wave comes, for example when it comes to ventilation, using solar shading and using heat-generating installations such as ovens and stoves.

It is important that the property owner works preventively with the health effects of heat waves, for example within their own control according to the environmental code (24). The Public Health Authority has general advice on indoor temperature that applies to both the property owner and the person running a business (25).

Consider whether a division of responsibility for handling heat can be clarified in the lease agreement with the property owner, as well as any requirements for preventive measures and preparedness measures.

Requirements in procurement and contracts with private contractors

An important measure could be to add special writings about heat waves in guidelines for procurement and agreements with care providers and private contractors. Requirements specifications and agreements can then clarify e.g. distribution of responsibilities, requirements that the self-control takes into account and manages heat, as well as preventive measures and emergency measures, etc.

Supervision and aid processing

The health protection supervision of various activities is carried out by the municipalities' environmental and health protection offices or the equivalent. Business here means, for example, a preschool, a school, a home for the elderly or an apartment building. Supervision means that the municipality checks that the law is followed and provides information and advice to those who run the business. The municipalities check that the activity does not pose an inconvenience to people's health. The supervision includes checking that there is protection against high temperatures (26).

In order to prevent high indoor temperatures, it is important that there is the possibility of solar shading, for example awnings and blinds. It is also important to have functioning ventilation and, if necessary, air conditioning. Such requirements for preventive measures can be set within the health protection supervision.

Aid officers also have a role in preventive work. They consult with care recipients, relatives and operations regarding the planning of the home, such as where the apartment is located in the property and available equipment for solar shading and cooling.

The heat also affects the staff in various operations and must therefore be handled as a work environment issue (27).

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Appendix 1. Example of a news release or press release

Six tips for better health during a heat wave

Climate change means that we can expect heat waves to become more common in Sweden as well. These periods of high temperatures can be stressful for health. The elderly and the chronically ill are among those who are particularly sensitive.

Sustained periods of heat during the summer can cause both health problems and increased mortality. This applies especially during the periods when the heat is high throughout the day. When the nights are also warm, the body does not have time to recover and it can be difficult to bring the temperature down indoors. **Mainly, it is people who are older, chronically ill or have certain types of disabilities who are particularly vulnerable. But even very young children and pregnant women are among the risk groups during a heat wave.** Preventive measures in healthcare and advice to the public can reduce ill-health rates and mortality caused by periods of high temperatures.

Here are six important pieces of advice for people who belong to the risk groups and their relatives¹ :

- 1. Pay attention** to the indoor temperature. The risk of health problems increases so soon the temperature rises indoors. The risks increase significantly when the temperature outside reaches up to 26 °C or more for three consecutive days.
- 2. Drink more.** Don't wait until you get thirsty. Eat liquid-rich foods such as vegetables and fruit. Avoid large amounts of sugary drinks and alcohol. Keep in mind that people close to you may need help with drinking.
- 3. Arrange a cool environment.** Use curtains, blinds and awnings. Try to be on the coolest place in the house. Air at night when it is cool.
- 4. Arrange cooling.** A cool shower is most effective. A wet towel around the neck is one alternative. Use loose-fitting clothes in natural materials, they are cooler than tight-fitting synthetic clothing.
- 5. Take it easy.** Avoid physical exertion during the hottest hours of the day.
- 6. Store medicines correctly.** Information is available if your medicine is temperature sensitive about storage on the packaging or in the package leaflet.

Warning signals for negative heat effects are elevated body temperature, pulse or breathing rate as well as new dizziness and abnormal fatigue. Dry mouth and reduced urine output can be signs of dehydration.

¹ When designing an internal news release, the advice should be changed to the target group-adapted advice to healthcare personnel.

Appendix 2. Examples of instructions for various functions in the alarm chain

The instructions in this appendix can be ordered as an editable Powerpoint file via info@folkhalsomyndigheten.se. Then enter the publication title and article number 21228.

There are five examples of instructions for different functions in the alarm chain:

- Instructions to contact point for alarm & coordination
- Instruction to administration/executive management for care and social care
- Instruction to operations manager & MAS
- Instruction for communication function
- Instruction to authorized personnel

Alarm chain during a heat wave

Instructions to contact point for alarm & coordination

When SMHI issues a message or yellow or orange warning about high temperatures, an alarm chain is activated in the organization. The alarm chain consists of a number of functions.

This is the instruction for the function of the alarm chain Contact point for alarms & coordination. The instructions describe four important steps and what is important to do in this particular function of the alarm chain.

Alarm & Coordination Point of Contact monitors the high temperature messages and warnings issued by SMHI.

MEDIATE HEAT WARNING

- The municipality's contact point for alarms and coordination has the task of alarming several functions in the alarm chain: certified healthcare staff, communication function or head of administration and MAS in case of administration for care and social care.
- Ask the next function if they know how to act, and possibly ask if they need advice or help with something.
- Remember to update the alarm if the level of the message or warning changes. The warning level may increase or decrease depending on the weather forecast and other factors. Please note that SMHI does not announce when the heat warning is over, so you must check the duration yourself via smhi.se or the SMHI weather app.
- Verbal direct contact is often most effective, but should ideally be combined with written contact information.

MEDIATE INFORMATION & COUNCIL

Forward information to:

- **Legitimate personnel.** Notify licensed healthcare personnel about SMHI's heat warning and refer to 1) alarm chain and instructions for various functions in the alarm chain, 2) information and training material and advice.
- **Communication function.** Report which heat warning that SMHI has issued, and provide a link to 1) alarm chain, 2) instruction for communication function, 3) informational and educational material and advice.
- **Administration/executive management** for care and social care. Report which heat warning that SMHI has issued, as well as providing a link to 1) alarm chain, 2) instructions for various functions in the alarm chain, and 3) informational and training materials and advice.

RECONNECT

- Check that the next function has been reached by your information. Repeat the alarm if it has not been successful. Please make personal contact.
- Exchange information with other functions in the alarm chain about how the heat development and the situation in general is progressing. Update each other on what consequences the heat has had, what needs for help have arisen, what coordination is lacking, what resource availability is possible, etc. This creates a common picture of the situation that should also be documented and taken forward in following up and improving the business's preparedness for heat wave.

DOCUMENT

- Document your contacts and activities. Feel free to use a template. The documentation is important to be able to follow up the work later and it lays an important foundation for new knowledge, conclusions and learning.

Alarm chain during a heat wave

Instruction for administration/executive management for care and social care

When SMHI issues a message or yellow or orange warning about high temperatures, an alarm chain is activated in the organization. The alarm chain consists of a number of functions.

This is the instruction for the function of the alarm chain Administration/executive management for care and social care. The instructions describe four important steps and what is important to do in this particular function of the alarm chain.

MEDIATE HEAT WARNING

- The administration for care and social care has the task of informing about the heat warning to the next function of the alarm chain "Operation Managers & MAS".
- It is important that measures are taken as soon as possible and not to delay information about risks, risk groups and possible measures.
- Remember to notify staff if the level of the message or warning changes.
The warning level may increase or decrease depending on the weather forecast and other factors. Please note that SMHI does not send out a message or warning that it is removed, but you must check the duration yourself.
- Verbal direct contact is often most effective, but should ideally be combined with written contact information.

MEDIATE INFORMATION & COUNCIL

- **Post advice in visible places.** Advice to various professional groups and to individuals and relatives is designed as grants.
- **Notify operational managers & MAS.** Report which heat warning that SMHI has issued, as well as provide links to 1) alarm chain, 2) instructions for various functions in the alarm chain, and 3) informational and educational materials and advice.

RECONNECT

- Check that the next function has been reached by your information. Repeat the alarm if it has not been successful. Please make personal contact.
- Exchange information with other functions in the alarm chain about how the heat development and the situation in general is progressing. Update each other on what consequences the heat has had, what needs for help have arisen, what coordination is lacking, what resource availability is possible, etc. This creates a common picture of the situation that should also be documented and taken forward in following up and improving the business's preparedness for heat wave.

DOCUMENT

- Document your contacts and activities. Feel free to use the same template for documentation to facilitate the compilation. The documentation is important to be able to follow up the work later and it lays an important foundation for new knowledge, conclusions and learning.

Alarm chain during a heat wave

Instruction to operations manager & MAS

When SMHI issues a message or yellow or orange warning about high temperatures, an alarm chain is activated in the organization. The alarm chain consists of a number of functions.

This is the instruction for the function of the alarm chain Operations Manager & MAS. The instructions describe four important steps and what is important to do in this particular function of the alarm chain.

MEDIATE HEAT WARNING

- Operations managers or MAS are tasked with informing about the heat warning for legi-temporary staff and other care staff including summer substitutes.
- It is important that measures are taken as soon as possible and not to delay the information about risks, risk groups and possible measures. Although authorized personnel may have already received the heat warning, for safety's sake the heat warning must be conveyed in several stages.
- Remember to notify staff if the level of the message or warning changes. Where-ning level may increase or decrease depending on the weather forecast and other factors. Please note that SMHI does not send out notices or warnings that are removed, but you must check the duration yourself.
- Verbal direct contact is often most effective, but should ideally be combined with written contact information.

MEDIATE INFORMATION & COUNCIL

- **Post the advice in visible places.** Advice to various professional groups and to individuals and relatives is designed as grants.
- **Legitimate personnel.** Notify the staff of the heat warning that SMHI has issued, and convey 1) the document Handling heat waves 2) advice for doctors, nurses and other licensed personnel and 3) link to the online training Handling heat waves in care and care.
- **Other care staff including summer substitutes.** Notify the staff of the heat warning that SMHI has issued, and pass on 1) the document Handling heat waves 2) advice for health and care staff in home care, home health care, special housing and primary care, and 3) link to the online training course Handling heat waves in care and care .

RECONNECT

- Check that the next function has been reached by your information. Repeat the alarm if it has not been successful. Please make personal contact.
- Exchange information with other functions in the alarm chain about how the heat development and the situation in general continues. Update each other on what consequences the heat has had, what needs for help have arisen, what coordination is missing, what resources are available, etc. This creates a common picture of the situation that should also be documented and taken forward in the follow-up and improvement of the business preparedness for heat wave.

DOCUMENT

- Document your contacts and activities. Feel free to use a template. The documentation is important to be able to follow up the work later and it lays an important foundation for new knowledge, conclusions and learning.

Alarm chain during a heat wave

Instruction for communication function

When SMHI issues a message or yellow or orange warning about high temperatures, an alarm chain is activated in the organization. The alarm chain consists of a number of functions.

This is the instruction for the alarm chain function Communication function, including on-call communicator. The instructions describe four important steps and what is important to do in this particular function of the alarm chain.

MEDIATE HEAT WARNING

- Spread message or warning about high temperatures.
- Remember to update the alarm if the message or warning changes.
The warning level may increase or decrease depending on the weather forecast and other factors. Please note that SMHI does not send out a message or warning that it is removed, but you must check the duration yourself.

MEDIATE INFORMATION & COUNCIL

- Produce and link to easily accessible and regular information in the channels that suit situation and target groups best.
- Channels can be the website, customer service, citizen offices, social media, radio announcements, press releases, etc.
- Colleagues who receive questions from outside, for example press service and customer service, can benefit from a compilation of questions and answers.
- Inform the idea-driven organizations that operate locally and meet risk groups and their relatives.
- Tell about places where citizens can seek to cool off, for example stone buildings or others public premises with air conditioning. Connect back to the function Contact point for alarms & coordination about which activities have been carried out and adjust the situational picture in the organization.
- Is there a special large outdoor event that needs information or can spread information?

RECONNECT

- Feedback to the function Contact point for alarms & coordination about which activities carried out and agree with the situation in the organization.
- Do any of the activities need to be repeated? Plan and prepare
- Do the target groups read the information? Analyze the web statistics and possibly make clarifications and updates.

DOCUMENT

- Document your contacts and activities. Feel free to use the same template for documentation to facilitate the compilation. The documentation is important to be able to follow up the work later and it lays an important foundation for new knowledge, conclusions and learning.

Communication in case of preparedness and risk

It is important that conflicting messages are not spread. There are many actors who communicate with citizens. It is therefore easy for rumors to spread and at the same time difficult for citizens to know where to turn for information and to know which authority is responsible for what and thus provides information..

All communication must therefore be:

- Easy to understand
- Honest
- Relevant
- Target group adapted
- Single
- Fast
- Correct
- Realistic
- Unique

Risk communication should give people answers to their questions about threats and risks and what they can do to prepare. Therefore, all risk communication needs to be planned and structured. The citizens' right to information must govern both the strategic and the message-wise tactical path choices and decisions in the crisis situation.

The information must answer:

- What is happening?
- Why is it dangerous?
- Symptom?
- Consequences for me?
- What should I do?
- Where and when is more information given?
- When is the danger over?

Alarm chain during a heat wave

Instruction for authorized personnel

When SMHI issues a message or yellow or orange warning about high temperatures, an alarm chain is activated in the organization. The alarm chain consists of a number of functions.

This is the instruction for the function of the alarm chain Authorized personnel. The instructions describe four important steps and what is important to do in this particular function of the alarm chain.

MEDIATE HEAT WARNING

- Legitimate staff have the task of informing about the heat warning to the function of the alarm chain "Other care staff incl. summer temps", as well as to "Operations manager & MAS" in the event that the authorized staff received the heat warning first.
- It is important that action be taken as soon as possible and not to delay the information about risks, risk groups and possible measures.
- Remember to notify staff if the level of the message or warning changes.
The warning level may increase or decrease depending on the weather forecast and other factors. Please note that SMHI does not send out notices or warnings that are removed, but you must check the duration yourself.
- Personal contacts are always more effective, combined with written information.

MEDIATE INFORMATION & COUNCIL

- **Notify operations manager and MAS.** Legitimate personnel may receive the warning first. Make sure to notify the operations manager and MAS about the heat warning and have a dialogue about what measures are to be taken.
- **Apply the advice.** Take part in 1) the publication Handling heat waves 2) advice for doctors, nurses and other licensed personnel and 3) the web training Handling heat waves in care and care.
- **Other care staff including summer substitutes.** Notify the healthcare staff of heat- the warning that SMHI has issued, and convey 1) the document Handling heat waves 2) advice for health and care staff in home care, home health care, special housing and primary care, and 3) link to the web training Handling heat waves in care and care.
- **Post the advice in visible places.** Advice to various professional groups and to individuals and relatives is designed as grants.

RECONNECT

- Check that the next function has been reached by your information. Repeat the alarm if it has not been successful. Please make personal contact.
- Exchange information with other functions in the alarm chain about how the heat development and the situation in general continues. Update each other on what consequences the heat has had, what needs for help have arisen, what coordination is lacking, what resource availability is possible, etc. This creates a common picture of the situation that should also be documented and taken forward in following up and improving the business's preparedness for heat wave.

DOCUMENT

- Document your contacts and activities. Feel free to use the same template for documentation to facilitate the compilation. The documentation is important to be able to follow up the work later and it lays an important foundation for new knowledge, conclusions and learning.

This guidance should provide support and recommendations for the work with developing action plans for prevention and management negative health effects of heat waves. The guidance is written for municipalities, regions and private actors who provide health care to the population and especially to the groups that are vulnerable to negative health effects of heat waves. The guidance provides examples of how you can work to develop action plans and highlights important aspects to consider.

A widespread heat wave can have serious consequences for society let and not least for the health of the population. Being exposed to heat however, entail different risks for different individuals. Risk groups are especially the elderly, the chronically ill, people with disabilities ning, young children, pregnant women and people taking certain medications.

As support in the work, the Public Health Agency has produced information material that contains advice adapted to the target group. The material can be downloaded from www.folkhalsomyndigheten.se/varmebolja

The Public Health Agency is a national knowledge authority that works for better public health. The authority does this by developing and support society's work in promoting health, preventing ill-health and protect against health threats.

Our vision is public health that strengthens society's development.



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